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FIFTH SEMESTER | B.Tech. (EE)

END TERM EXAMINATION | Nov-2022

EE-301 POWER ELECTRONICS

Time: 3:00 Hours | *Max. Marks: 40*

Note: Answer ANY EIGHT questions and all carry equal marks. Write the ANSWERS IN A SEQUENCE. Assume any data is missing and clearly mention the assumption.

Q(1) What do you mean by two power transistor model of Silicon Controlled Rectifier? Prove that the anode current of the Silicon Controlled Rectifier is given by: $I_A = \frac{\alpha_2 I_G + I_{CBO1} + I_{CBO2}}{1-(\alpha_1+\alpha_2)}$. Discuss the following terms of SCR (a) Finger voltage (b) Latching current (c) Holding current (d) Surge Current

[3+2=5M][CO1]

Q(2) Draw the symbol and V-I characteristics of the following devices: (a) Power Diode (b) Power BJT (c) SCR (d) TRIAC (e) GTO

[5x1=5M][CO1]

Q(3) Derive the expression for the output voltage of a single phase half wave controlled converter feeding R-L load with freewheeling diode while considering the firing angle α? Also determine the DC output voltage, in case the same circuit as shown in Fig.1 is energized with 120V, 50Hz supply while assuming the firing angle $\alpha = 30°$. $R = 5\Omega$, $L = 0.1H$.

[3+2=5M] [CO2]

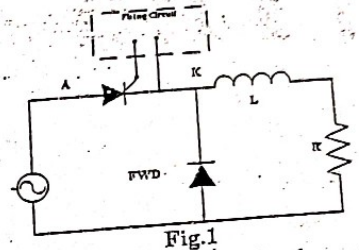


Fig.1

Q(4) Explain the working principle of dual converter with the help of suitable waveforms. A single-phase dual converter is supplied from a 220 V, 50 Hz ac source and it is connected to a load of $R = 20\ \Omega$. The value of the current limiting reactor is L = 0.08H. If the firing angle o converter-1 $\alpha_1 = 45°$, determine (a) the firing angle of converter-2 (b the Average value of load voltage.

[3+2=5M][CO3

Q(5) Discuss the analytical and graphical analysis of a three-phase full-bridge converter having a source inductance L_s in each line with RL load by assuming that the load current is constant and ripple free. A six-pulse three-phase full-bridge converter has a delay angle of $\alpha = 120^\circ$ and operates in an inverting mode. The three-phase ac system is 4160 V RMS line-to-line. The dc source is 3000 V, R=2Ω, and load inductance L=0.15H is large enough to consider the current to be purely dc. Determine the power transferred to the ac source from the dc source.

[3+2=5M][CO3]

Q(6) Discuss the analytical and graphical analysis of a single-phase bidirectional ac voltage controller with RL load. An AC load circuit consists of a resistance of 30Ω in series with an inductance of 0.2H. it is fed from a 120V, 50Hz AC source. Determine the firing angle range in which the load voltage can be controlled and the conduction interval in a half cycle for a firing angle delay of 100°. [3+2=5M][CO4]

Q(7) Explain the operating principle of mid-point-type single-phase step-up cyclo-converter. Mention the conduction of various thyristors in the waveforms. [5M][CO4]

Q(8) Explain the operating principle of the step-down chopper with the help of voltage and current waveforms. Also derive expressions for average output voltage and RMS output voltage. A chopper-controlled electric train is powered by a 1500V DC supply. The power semiconductor switching element has a minimum effective ON time of 40μs. During starting and slow-speed running the output of the chopper has to go as low as 15V. What is the highest chopper frequency possible to satisfy this requirement?

[3+2=5M][CO5]

Q(9) Give a brief description for the operating principle of a basic series inverter with help of analytical and graphical analysis. Discuss Resistance triggering of SCR and the conditions for forced commutation along with Class-A commutation of SCR.

[2+1+2=5M][CO5]

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FIFTH SEMESTER

B.TECH. (EE)

END SEM EXAMINATION

Nov.-Dec 2022

EE-303 POWER TRANSMISSION & DISTRIBUTION

Time: 3:00Hours

Max. Marks : 40

Note: Answer any FIVE questions. Assume suitable missing data, if any.

Q1 Answer briefly: 2x4=8

i) Draw the single line diagram of a typical power system. Show various voltage levels also. [CO1]

ii) A 100km long line has a characteristic impedance of 350 ohms. What will be the characteristic impedance if the length of line is doubled? Give reason in support of your answer. [CO3]

iii) Explain why the voltage is unequally distributed across the insulators in a string of insulators. [CO5]

iv) A single phase overhead line 40km long consists of two parallel wires of 6mm radius, 1.5m apart. If the system voltage is 50kV at 50Hz, calculate the capacitance and charging current with the line open circuited. [CO2]

Q2 a. Define 'Characteristic Impedance' and 'Propagation Constant' for a transmission line. Also explain their significance. [CO3] 2

b. A 3-phase 50Hz 200km long transmission line has following distributed parameters per km: r=0.2ohms, x=0.41ohms, y =3.14x10^{-6} mho. The sending end voltage is 220kV. Determine A, B, C, D parameters of the line and hence determine the sending end current and receiving end voltage of the line under open circuit condition. Consider the line as long line. [CO2,3] 6

Q3 a. Derive the expressions for sag and tension in an overhead transmission line conductor when the supports are at unequal heights. [CO5] 4

b. An overhead line is supported by towers at the same level having a span of 250meters. Calculate the sag in i) in still air condition with no ice

covering ii) under a wind pressure of 400N/m² and ice covering of 1 cm thickness. The conductor diameter and weight are 1.1cm and 650kg/km, respectively. The breaking load and factor of safety are 7000kg and 2 respectively. Ice weighs 915kg/m³. [CO5] 4

Q4 a. Define string efficiency of a string of overhead line insulators. Explain the methods to improve it. [CO5] 4

b. In a string of three disc units, capacitance of each link pin to earth is 10% of the capacitance of one unit. If a voltage of 33kV, 50Hz is applied across the string, calculate i) voltage across each unit ii) string efficiency. [CO5] 4

Q5 a. For a given voltage and sheath diameter, deduce the size of conductor of a single core cable which gives minimum gradient at the surface of the conductor. Also explain the condition for stable operation of the cable. [CO4] 4

b. A 60kV single core cable has a conductor diameter of 2.5 cm and sheath of inside dielectric 6cm. Calculate the maximum stress. It is desired to reduce the stress by using two intersheaths. Determine their best positions, maximum stress and voltage on each. [CO4] 4

Q6 a. Explain different types of distribution systems for the supply of electrical power. [CO1] 4

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b. A three phase 220kV, 50Hz transmission line has conductors spaced 5.5meters at the corners of an equilateral triangle. Calculate the corona inception voltage if each conductor has a radius of 1.2cm with surface irregularity factor of 0.92. The temperature of the surroundings is 40°C and atmospheric pressure is 752mm of mercury. Also find the corona loss/km length of the line if there is an over voltage of 1.6pu. [CO3]4

Q7. Write short notes on any TWO 4x2 = 8

i. Lightning Arrestor for protection against over voltages [CO3]

ii. HVDC Transmission [CO1]

iii. Determination of A, B, C, D parameters of a transmission line in lab [CO2]

iv. Determination of capacitance of a 3-core cable [CO4]

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5th and 7th SEMESTER

B.Tech

END TERM EXAMINATION *Nov-2022*

EE319 Digital System Design

Time: 3:00 Hours *Max. Marks: 40*

Note :
If necessary, make appropriate assumptions and approximations and state them
You must solve all the questions and their parts in the serial order

Q.1 Answer any FIVE. [2X5=10]

(a) What are the major limitations associated with Static CMOS logic? [CO#2]

(b) Explain why present VLSI circuits use MOSFETs instead of BJTs? [CO#1]

(c) What is Verilog? How is it different from normal programming languages? [CO#4]

(d) "NMOS transistor cannot pass strong logic 1". Justify the statement. [CO#2]

(e) Write subthreshold current equation for n-MOSFET. [CO#1]

(f) Why TSPC latches are preferred as compare to Static D latches? [CO#3]

Q2. Answer any THREE. [3X3=9]

(a) Sketch a pseudo-nMOS gate that implements the function

$$F = \overline{A(B+C+D)+E\cdot F\cdot G}$$ [CO#2]

(b) Design a 2 input XOR gate using transmission gate logic and write a Verilog module for the circuit designed. [CO#4]

(c) Differentiate the between the process of ion implantation and diffusion in fabrication of a MOS transistor. [CO#1]

(d) What is the function of the circuit in Figure 1? [CO#3]

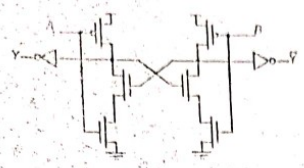


Figure 1

Q3. Figure 2 contains a pass-gate logic network. [3] [CO#2,3]

(a) Determine the truth table for the circuit. What logic function does it implement?

(b) If the PMOS were removed, would the circuit still function correctly? Does the PMOS transistor serve any useful purpose?

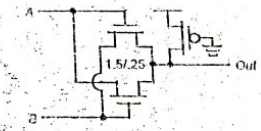


Figure 2

Q4. Sketch a 3-input XOR functions using each of the following circuit techniques: [3][CO#2]

(a) Static CMOS

(b) Pseudo-nMOS

Q5. Design and explain the working of TSPC Flip-Flop. [2] [CO#3]

Q6. Ignore all capacitances at the "out" node: [5][CO#2]

(a) In the high to low transition in Figure 3, which transistor would first turn from ON to OFF state?

(b) During the high to low transition in Figure 3. What are the regions of operation of transistors Mp1 – Mp3 and Mn1 – Mn3 for Vin = 0.7V (Vth = 0.3V)?

(c) What is the value of the current through PUN?

(d) During the low to high transition, which of Mn1 and Mn2 first turns ON?

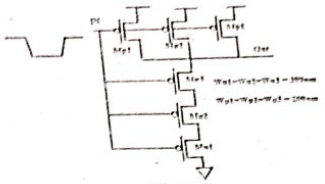


Figure 3

Q7. Consider the circuit of Figure 4. Assume the inverter switches ideally at VDD/2, neglect body effect, channel length modulation and all parasitic capacitance throughout this problem. [4] [CO#2]

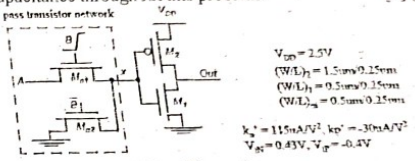


Figure 4

(a) What is the logic function performed by this circuit?

(b) Explain why this circuit has non-zero static dissipation.

(c) Using only just 1 transistor, design a fix so that there will not be any static power dissipation.

(d) Implement the same circuit using transmission gates.

Q8. Write a Verilog code for Full Adder Circuit with test bench. [4] [CO#4]

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V & VII SEMESTER

B.Tech

END TERM EXAMINATION *Nov/Dec-2022*

EE323 FUNDAMENTALS OF MACHINE LEARNING

Time: 03:00 Hours *Max. Marks: 40*

Note : All questions carry equal marks (8 marks).
Attempt "ANY 5" questions
Assume suitable missing data, if any.

Q.1 For each statement below, say whether it is true or false, and give a one or two sentence justification of your answer. [8][CO2,3,5]

A. As we increase the number of parameters in a model, both the bias and variance of our predictions should decrease since we can better fit the data.

B. Machine learning models work automatically without the need for the user to manually set hyperparameters.

C. The k-means algorithm is guaranteed to converge to the global minimum.

D. When optimizing a regression problem where we know that only some of our features are useful, we should use $L1$ regularization.

Q.2 ***[Feature Engineering and Evaluation Metrics]*** [8][CO4]

A. Why is feature normalization necessary in Machine Learning? Suggest some techniques?

B. Suppose 10,000 patients get tested for flu; out of them, 9000 are actually healthy and 1000 are actually sick. For the sick people a test was positive for 620 and negative for 380. For the healthy people, the same test was positive for 180 and negative for 8820. Construct a confusion matrix for the data and compute the accuracy, precision and recall for the data.

Q.3 **[Gradient Descent Algorithm]** [8][CO3]

Run the Gradient Descent algorithm on the error function $E(w_0, w_1) = \frac{1}{2}w_0^2 + w_1^2$ starting at the point $W^{(0)} = [2,2]^T$ and using step size of $\frac{1}{2}$. Output the weight parameters $W^{(1)}$, $W^{(2)}$, $W^{(3)}$, where W vector corresponds to $[w_0, w_1]^T$.

Q.4 **[Principal Component Analysis]** [8][CO4,5]

A. Explain the method of Principal Component Analysis (PCA) in machine learning.

B. Given 3 2D data points: (0,0), (1,1), (-1, -1). Do the following:

i) Find the 1st principal component without solving for eigenvectors or SVD and justify.

ii) Project the original data onto the 1st principal component axis and find the transformed coordinate.

Q.5 **[Clustering]** [8][CO5]

A. The k-means algorithm is designed to reduce the mean squared error between the cluster centers and their assigned points. Will increasing the value of k ever increase this error term? Why or why not?

B. Use K Means clustering to cluster the following data into two groups. Assume cluster centroid are C_1=2 and C_2=4. The distance function used is Euclidean distance.
Data: { 2, 4, 10, 12, 3, 20, 30, 11, 25 }

Q.6 **[Support Vector Machines]** [8][CO6]

A. Explain with a suitable dataset of your choice and plots, how Support Vector Machine can be used for classification of linearly separable data.

B. Describe the significance of Kernel functions in SVM. List any two kernel functions.

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V - SEMESTER

B.Tech

END TERM EXAMINATION Nov/Dec-2022

HU 301 - ENGINEERING ECONOMICS

Time: 03:00 Hours

Max. Marks: 50

Note: Attempt any five questions. All questions carry equal marks.

Q.1 (a) A consumer consumes only two goods X&Y both priced at 3 per unit. If the consumer chooses a combination of these two goods with the marginal rate of substitution equal to 3. Is the consumer in equilibrium? Give reasons. What will a rational consumer do in this situation? [CO:1][5 Marks]

(b) What do you mean by Production Function? Explain the likely behaviour of Output when all the inputs are increased in the production process. [CO:1,2][5 Marks]

Q.2 (a) How are commercial banks different from the Central Bank? Discuss the mechanism of credit creation by commercial banks. [CO:2][5 Marks]

(b) What causes the fluctuations in the Economic output over a period of time. Discuss phases of the business cycle. [CO:2,3][5 Marks]

Q.3 (a) Investment A costs Rs.10,000 today and pays back Rs.11,500 two years from now. Investment B costs Rs.8000 today and pays back Rs.4500 each year for two years. If an interest rate of 5 % is used, which alternative is superior? [CO:4] [5Marks]

(b) A company can make a particular component or purchase from the market. The cost detail is as below:

If it purchases from the market. The purchasing price per unit	Rs.3050+GST @18%
If it makes:	
Cost of the machine	Rs 1020300+GST @18%

Salary of the machine operator per month	Rs 30000
Rent of the workshop per month	Rs25000
Raw material-1 per unit	Rs 400+GST@18%
Raw material -2 per unit	Rs 1100 +GST @18%
Other cost per unit	Rs 300

If the annual demand is 1500

Whether company should 'Make' or 'purchase'. Suggest [CO:2,4] [5 Marks]

Q.4 (a) A monopolist has the cost function $TC(y) = 200y + 15y^2$ and faces the demand function given by $p = 1200 - 10y$. On the basis of given information, calculate the following: [CO:1,4][5 Marks]

(i) What output maximizes its profit?
(ii) What is the profit-maximizing price?
(iii) What is its maximal profit?

(b) Discuss the guiding principles of Indian Five Year Plans. [CO:3][5 Marks]

Q.5 (a) A person is planning for his retired life. He has 10 more years of service. He would like to deposit 20% of his salary which is 4000 at the end of the first year and thereafter he wishes to deposit the amount with an annual increase of Rs 500 for the next 9 years with an interest rate of 15%. Find the total amount at the end of the 10th year of the above series. [CO:4][5 Marks]

(b) What do you mean by Fiscal policy? How can Fiscal Policy be used to close an Inflationary and Recessionary gap? [CO:2,3][5 Marks]

Q.6 (a) A company invests in one of the two mutually exclusive alternatives. The life of both alternatives is estimated to be 5 years with the following investments, annual returns and salvage value.

	Alternatives	
	A	B
Investment (Rs)	-1,50,000	-1,75,000
Annual equal return (Rs)	+ 60,000	+70,000
Salvage value (Rs)	+15,000	+35,000

Determine the best alternative based on the annual equivalent method if i = 25 % [CO:4] [5 Marks]

(b) Discuss the major trade barriers imposed by countries to restrict trade. [CO:2,3] [5 Marks]

Q.7 (a) A company is planning to expand its business after 10 years. To meet the expansion expenditure, at the end of first year the company is planning to deposit Rs. 20,00,000 in the reserve and from the next year it will increase the amount to be deposited Rs. 15,000 from the previous deposit for the next 9 years with an interest rate of 12%. Calculate the total amount which the company will have for the expansion at the end of 10 years. [CO:4] [5 Marks]

(b) From the following information calculate the break- even point and the turnover (sales) required to earn a profit of Rs 36000.

Fixed overheads	Rs. 18000
Variable cost per unit	Rs. 2
Selling price	Rs. 20

If the company is earning a profit of Rs. 36000 express the margin of safety level. [CO:4][5 Marks]